

Class 12 Timer Interrupt Class Program

```
/*  
 * File:  main.c  
 * Author:  Lenovo  
 *  
 * Created on 19 June, 2025, 10:12 AM  
 */  
  
#include <xc.h>  
  
#define _XTAL_FREQ 6000000  
  
#pragma config FOSC = HS    // High-speed external oscillator (4-20 MHz crystal)  
#pragma config WDTE = OFF   // Watchdog Timer Enable (WDT disabled)  
#pragma config PWRTE = ON   // Power-up Timer Enable (enabled)  
#pragma config BOREN = ON   // Brown-out Reset Enable (enabled)  
#pragma config LVP = OFF    // Low-Voltage Programming (disabled)  
#pragma config CPD = OFF    // Data EEPROM Memory Code Protection (disabled)  
#pragma config WRT = OFF    // Flash Program Memory Write Enable (off)  
#pragma config CP = OFF     // Flash Program Memory Code Protection (disabled)  
  
void TimerInit(void);  
  
int count;
```

```

void main(void)
{
    TimerInit();

    while (1)
    {
        PORTC = 0x00;

        __delay_ms(500);

        PORTC = 0xFF;

        __delay_ms(500);
    }
}

// Timer1 Initialization
void TimerInit(void)
{
    // Set PORTC and PORTD as output
    TRISC = 0x00;
    TRISD = 0x00;
    PORTC = 0x00;
    PORTD = 0x00;

    // Timer1: Prescaler 1:1, oscillator off, Timer1 on
    T1CON = 0x01; // 0000 0001

    /* TIMER INTERRUPT CONFIGURATION */
    // Enable Global and Peripheral Interrupts
    INTCON |= (0x01 << 7); // GIE
    INTCON |= (0x01 << 6); // PEIE

```

```

// Enable Timer1 interrupt
PIE1 |= (0x01 << 0); // TMR1IE

// Clear Timer1 interrupt flag
PIR1 &= ~(0x01 << 0); // TMR1IF
}

// Interrupt Service Routine
void __interrupt() ISR(void)
{
    if (PIR1 & (0x01 << 0)) // TMR1IF
    {
        count++;
        if(count == 22)
        {
            // Toggle PORTD on Timer1 interrupt
            PORTD = ~PORTD;
            count = 0;
        }

        // Clear Timer1 interrupt flag
        PIR1 &= ~(0x01 << 0); // Clear TMR1IF
    }
}

```